BIT 4543 — ARTIFICIAL INTELLIGENCE GROUP PROJECT

AI Sentiment Analysis Dashboard

Classifying Amazon customer reviews with TF-IDF and Logistic Regression

Positive

Neutral

Negative

PROJECT TEAM

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Background & Problem Statement

Online shopping has led to a huge number of customer reviews being published on the Web. However, it becomes impractical to read each and every one of them to understand the overall sentiment and opinion.

Reviews also come in messy with informal language and different types of errors like wrong spellings, HTML code, and sentiment words that change directions in the same sentence.

Besides, the major dilemma is in selecting a method that not only performs well but does it without the burden in terms of the deep learning or the transformer model cost and complexity.

So: can a lightweight TF-IDF + Logistic Regression pipeline classify Amazon reviews well enough to power a real, interactive dashboard?

SCALE OF THE CHALLENGE

2

Product categories analysed

100K

Reviews sampled (50K per category)

3

Sentiment classes: negative, neutral, positive

PURPOSE

Aim & Objectives

Aim: Develop and evaluate a machine-learning system that automatically classifies Amazon reviews as positive, neutral, or negative.

1. Build an AI app that classifies review text into 3 sentiment classes
2. Collect & clean a large sample from Video Games and All Beauty categories
3. Train category-specific TF-IDF + Logistic Regression models
4. Visualise sentiment distributions, ratings, and model performance
5. Create a real-world interface while allowing users to optionally provide feedback.
6. Evaluate with classification metrics and user acceptance testing

Project Scope

Sentiment labels are derived directly from star ratings:

1–2 ★	3 ★	4–5 ★
Negative	Neutral	Positive

In scope

- Category selection & review prediction
- Probability display for each class
- Dataset charts & model metrics
- Analysis history & feedback recording

Out of scope

- User login & real time Amazon data collection
- Aspect based sentiment explanation
- Multilingual modelling
- Automated business decision making, file upload

Literature Review Summary

Study	Domain	Approach	Relevance
Adak et al. (2022)	Food-delivery reviews	ML, DL & XAI review	Background on approaches & explainability
Chinnalagu & Durairaj (2021)	Customer reviews	Linear ML, FastText, neural	Supports linear models for text classification
Hashmi & Yayilgan (2024)	Amazon product reviews	TF-IDF, Word2Vec, FastText, ML/DL/transformers	Directly relevant Amazon review sentiment
Shaik Vadla et al. (2024)	Amazon reviews	BERT/T5, aspect-based	Demonstrates transformer approaches

Research gap: Advanced deep-learning and transformer approaches can boost accuracy, but need heavier compute and implementation effort. This project targets a practical, interpretable middle ground TF-IDF + Logistic Regression evaluated experimentally against simpler baselines.

APPROACH

End-to-End Methodology

Amazon Reviews'23 → Balanced 50K Sample → Rating → Sentiment Label → Text Preprocessing → TF-IDF Features → Train/Test Split (80/20) → Model Training → Best Model + Streamlit App

Same workflow applied independently to both Video Games and All Beauty categories, so each gets its own tuned model.

80/20

stratified train/test split

30,000

TF-IDF features (uni + bigrams)

Logistic Regression

final selected classifier

Dataset & Preprocessing Yield

Video Games

48,371

final reviews used for modelling
(from 50,000 sampled)

All Beauty

48,443

final reviews used for modelling
(from 50,000 sampled)

Video Games cleaning pipeline

- Sampled: 50,000
- – missing text: 49,997
- – duplicates: 48,476
- – empty after cleaning: 48,375

WHY PRESERVE NEGATION?

"not good" should never be reduced to just "good".

Preserved words: not, no, never, nor, neither, hardly, barely, nothing kept through stop word removal and lemmatization.

Text Preprocessing Pipeline

1. Remove missing / duplicate reviews
2. Lowercase all text
3. Strip HTML tags & URLs
4. Remove punctuation & numbers
5. Collapse excess whitespace
6. Remove English stop words
7. Preserve negation terms
8. WordNet lemmatization
9. Drop reviews empty after cleaning

TF-IDF Feature Extraction

Parameter	Value
Maximum features	30,000
N-gram range	1–2 (unigrams + bigrams)
Min document frequency	2
Max document frequency	0.95
Sublinear TF	Enabled

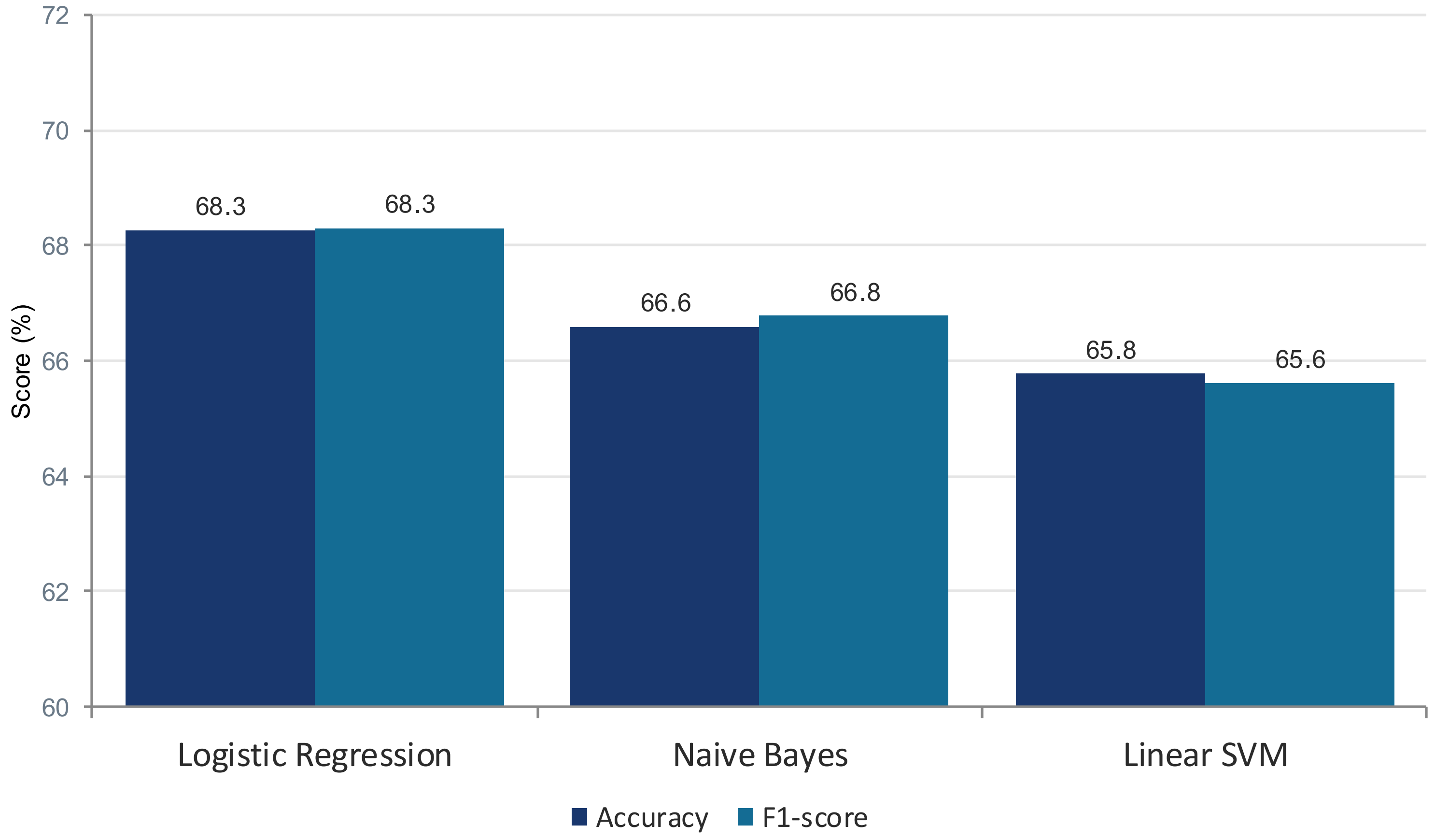
N-GRAMS CAPTURE MORE MEANING

- "good"
- "very good"
- "not good"
- "great game"

TF-IDF was fit only on training data, then applied to the test set preventing test data leakage into the vocabulary.

Comparing Classification Models

Video Games dataset Accuracy vs. F1-score



WHY LOGISTIC REGRESSION

- Highest accuracy & F1 across both categories
- Handles high-dimensional sparse TF-IDF features efficiently
- Outputs class probabilities, not just a label
- Simple, fast to train, easy to deploy

Final Model Performance

Video Games

68.34%
Accuracy

68.37%
Weighted F1

All Beauty

69.33%
Accuracy

69.28%
Weighted F1

Video Games perclass breakdown

Sentiment	Precision	Recall	F1-score
Negative	0.70	0.72	0.71
Neutral	0.58	0.59	0.58
Positive	0.77	0.74	0.76

Neutral is hardest to classify it overlaps with weak positive and negative expressions, pulling down its precision and recall.

Confusion Matrix Video Games

Actual (rows) vs. Predicted (columns)

	Negative	Neutral	Positive
Negative	**2,389**	760	150
Neutral	822	**1,913**	529
Positive	200	609	**2,303**

Reading it: diagonal cells (bold) are correct predictions. Most confusion happens between Neutral and its neighbours 822 neutral reviews were predicted negative and 529 were predicted positive.

The Streamlit Dashboard

Core features

- Select product category
- Enter a review and get an instant sentiment prediction
- View class probabilities
- Explore processed dataset charts & model metrics
- Browse analysis history and submit feedback

EXAMPLE PREDICTION

"Absolutely love this best purchase all year!"

Positive	96.13%
Neutral	3.46%
Negative	0.41%

Deployed on Streamlit Community Cloud, connected to GitHub updates to the main branch trigger redeployment automatically.

User Acceptance Testing

Tester	Tasks Passed	Tasks Failed	Status
Tester 1	8/9	1	Fail/Partial
Tester 2	8/9	1	Fail/Partial
Tester 3	9/9	0	Pass

Tasks covered: open app, select category, submit positive/negative/blank reviews, switch category, open dashboard & history, submit feedback.

KEY OBSERVATIONS FROM TESTING

- Long, mixed reviews sometimes misclassified as neutral
- Sarcasm not reliably understood read as positive
- Slang / abbreviations occasionally missed
- Core flow predict, switch category, dashboard, history, feedback worked smoothly for all testers

Limitations & Future Improvements

LIMITATIONS

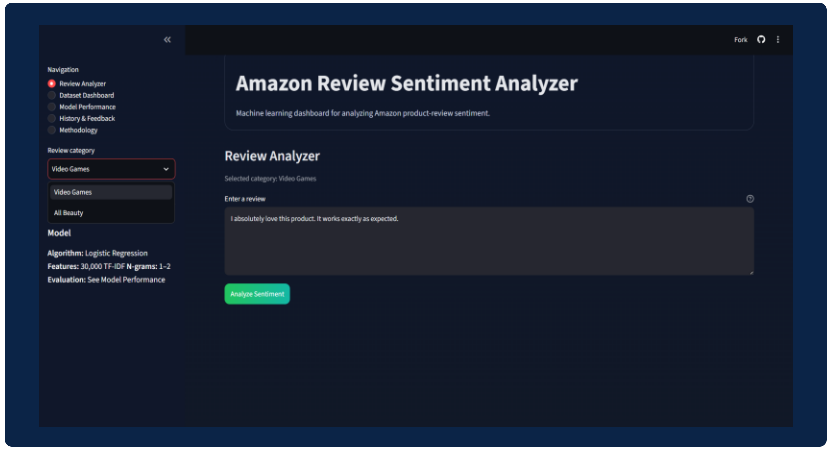
- Labels inferred from star ratings, not manual annotation
- English language reviews only
- TF-IDF has limited contextual understanding struggles with sarcasm & mixed sentiment
- Results based on a single 80/20 split
- Model may not generalise well to unseen product categories

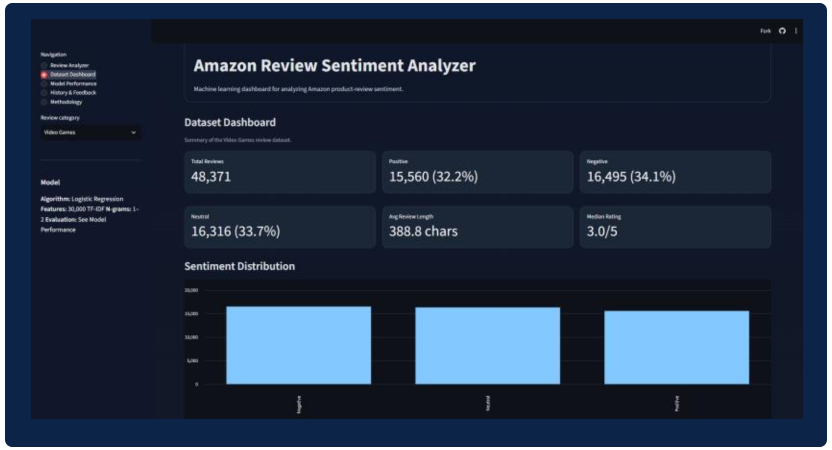
FUTURE IMPROVEMENTS

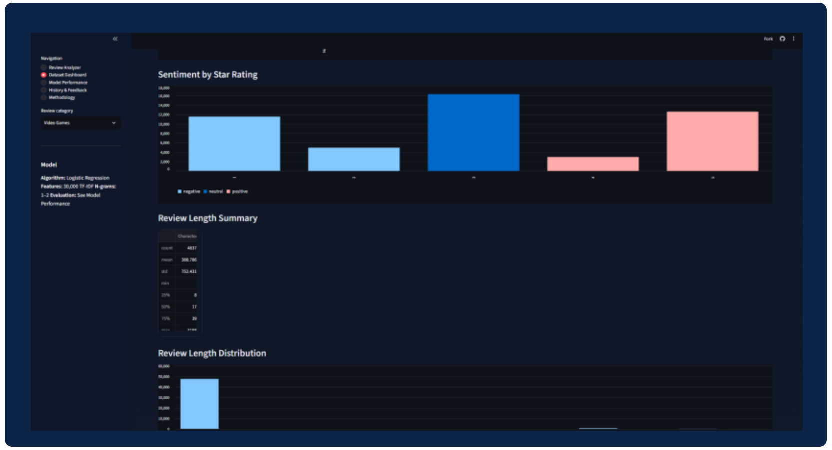
- Transformer models BERT, RoBERTa for richer context
- Aspect-based sentiment analysis graphics, price
- Explainability via LIME or SHAP
- Larger, more diverse datasets and categories
- Cross-validation and deeper error analysis

APPLICATION

SCREENSHOT OF APP







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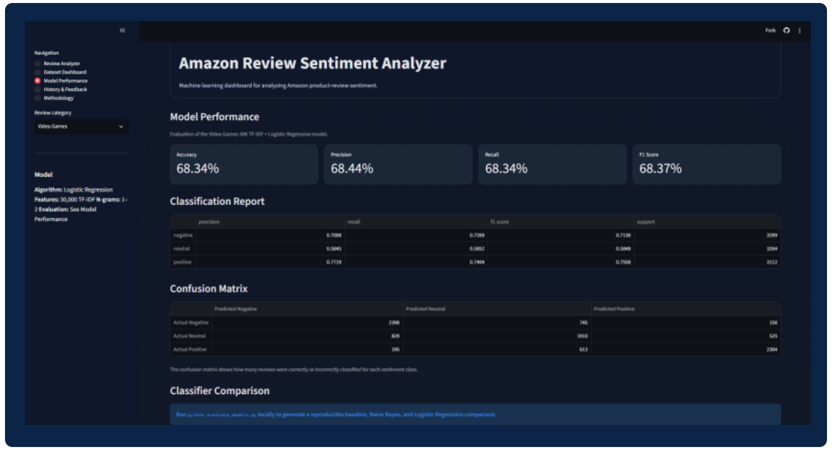

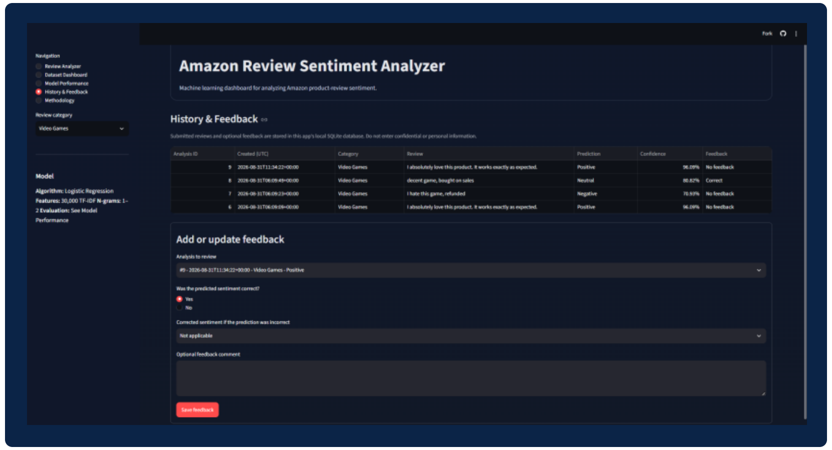

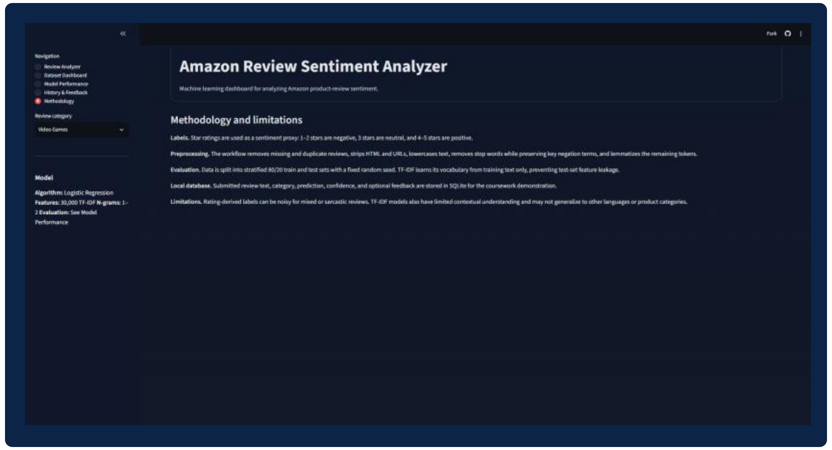

CONCLUSION

A Practical Approach That Works

TF-IDF + Logistic Regression delivers a reproducible, interpretable, and deployable sentiment-analysis pipeline without the cost of deep learning or transformer models.

68.34%

Video Games accuracy

69.33%

All Beauty accuracy

68.37%

Video Games F1-score

3/3

Testers rated the app usable

Traditional machine learning, applied thoughtfully, still delivers real practical value.

Thank you

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